Easy Network: A Framework Bridging Structured Data Management, Network Analysis and Interactive Visualization

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1. Introduction

Since the latter half of the last century, network analysis methods have gradually matured and expanded into the social sciences and humanities. Although it is still young and rising, the practice and application of network analysis methods in archaeology is becoming increasingly sophisticated and systematic, and the specific approaches to applying these methods in archaeological contexts are growing more complex and diverse. On the other hand, the supporting software designed to streamline the process of archaeological network analysis are not yet sufficiently developed, particularly during data processing, filtering, and final visualization including the interactivity and effectively flatten multidimensional data on network graphs. Today, researchers can process data using Python and R, and visualize it using Python's dedicated package NetworkX or the independent software Gephi, a modular and graphical network analysis tool. There remains a technical gap between data analysis and visualization, requiring to be handled separately and often rely on interdisciplinary collaboration or archaeologists to possess a certain level of statistical and computational expertise. The technical barriers might reduce research efficiency, forcing scholars to expend more effort, time, and energy on interdisciplinary conversation or learning specific programming skills, thereby diverting their attention to some extent away from exploring the full potential of Network analysis itself.

Facing these challenge, I attempt to develop a website-based tool Easy Network that integrates data management, filtering, network analysis and interactive visualization. It is designed to be user-friendly, featuring structured data organization and management, synchronization of network graphs with the database, interactive nodes and edges that can be easily traced back to the database, and support for collaborative data management and visualization among multiple users.

This essay will introduce the background of this tool’s development, including its theoretical framework, archaeological applications, statistical and technical support, and the assumptions underlying its quantitative research and data organization methods. It will also explore the architecture, specific features, user guide within the tool itself.

2. Theory Framework

Data management is neither merely a technical process of data storage nor neutral "container," but also a form of research infrastructure and a means of organizing knowledge and constructing research objects. In this chapter, I attempt to discuss the impact of data management as infrastructure on researchers and their work from a data science perspective, as well as the importance of consciously organizing structured data to serve research objectives.

2.1. Data Management as Research Infrastructure

Manovich (2001, pp. 212-225) proposed the concept of "Database as Symbolic Form," arguing that databases not only store information but also determine how information is organized, classified, and linked. Therefore, databases themselves embody researchers' understanding of materials and their organizational logic, rather than merely reflecting objective reality. This perspective resonates with Bowker and Star's discussion of classification systems. They view classification systems as an "invisible infrastructure" that profoundly influences the production of knowledge and the construction of social reality (Bowker & Star, 1999, pp. 1-10).

The very way data is organized embodies preconceived understandings and shapes how future researchers and readers interpret it. Even though many studies including those in the humanities and social sciences, which have increasingly adopted quantitative approaches in recent years--adhere to the FAIR principles (Findable, Accessible, Interoperable and Reusable) (Wilkinson et al., 2016) in their use and presentation of data, the specific methods and processes researchers employ to organize data are not always fully traceable. This creates barriers for readers without relevant background knowledge who wish to verify the validity of such data. In other words, even if the "FAIR principles" are implemented in the workflow, differences in training backgrounds can undermine the practical effectiveness of these principles when it comes to "effectively reprocessing data and reproducing results." While it may not be possible to completely bridge the gaps between disciplines and individual backgrounds, this essay aims to remind researchers that, in addition to the importance of publishing raw data and presenting results, the aspects involving human "manipulation" in the process are also of critical importance. Out of the same objective, the archaeological network analysis tool Easy Network is developed, which will be discussed later in details.

2.2. Network Analysis

Network analysis is a wide set of concepts, methods, and tools used to explore, study, and interpret the connections among entities, including objects, institutions, and individual humans. This theory focuses on the relationships between different entities at the macro level and uses a network graph to visualize these relationships, helpful for capturing richer information that may be overlooked due to excessive focus on specific individuals at the micro level. The entities are usually abstracted as "nodes" or "vertices," and the metaphorical or data-based links between the nodes are abstracted as "edges" or "links." (Newman, 2010, pp. 211-219; Brughmans, 2013, p. 624-626; Filet & Rossi, 2023, p. 15-16).

The methodology originated from graph theory, which was established by Euler (1736), and was initially used to study relational structures composed of nodes and edges. In the mid-20th century, with the development of sociology and anthropology, researchers began applying graph theory to the study of social relationships, gradually giving rise to Social Network Analysis (SNA), which focuses on interaction patterns among individuals, groups, and organizations and their social significance (Wasserman & Faust, 1994). At the end of the 20th century, the emergence of Complex Network Theory further propelled the development of network analysis, with models such as Small-World Networks and Scale-Free Networks expanding its scope of application in the study of complex systems (Watts & Strogatz, 1998; Barabási & Albert, 1999). Since the beginning of the 21st century, network analysis has gradually matured and gained widespread application in the humanities, including history, archaeology, and anthropology. Researchers have utilized network analysis to explore topics such as trade exchanges, social interactions, knowledge dissemination, migration processes, and regional connections, thereby revealing complex relational structures and dynamic interactive processes that are difficult to identify using traditional research methods (Brughmans, 2013; Knappett, 2013).

In an archaeological context, nodes can represent artifacts, people, institutions, or sites; edges represent connections, including communication, affiliation, or similarity (Filet & Rossi, 2023, pp. 15–16). Depending on the definition of “relationship” or “association,” network analysis can be employed to explore archaeological topics from diverse perspectives. Through either theoretical or data modelling, a wider and deeper view of the research topic can be presented, sometimes inspiring new ways of interpretation and thinking (Filet & Rossi, 2023, p. 17-19). In some cases, the edges can be assigned meanings, such as categorical labels or metrical weights, to more precisely describe and distinguish different edges, for instance, the strong or weak links. The combination of network analysis and classical methods could reveal information that is hardly delivered as text and by qualitative data. Nevertheless, network analysis is only a process of modelling and reorganizing what has already existed instead of generating new data or materials. It is always necessary to rely on solid data, classical methods, and contextual conditions, within which interpretation should be situated under a specific disciplinary background.

2.3. SQL Databases and Network Thinking

The development of modern databases has further reinforced a research paradigm centered on "relations." Before the advent of the relational model, most database systems primarily organized data using tree structures and network models. These systems typically required users to understand specific data access paths, and program logic was tightly coupled with data storage structures; consequently, any change in the underlying structure often necessitated a redesign of the application (Codd, 1970, pp. 377–378). To address this issue, Codd (1970) proposed the Relational Model, advocating for the organization of data through "relations" rather than physical storage paths. In this thinking, data exists in the form of tables, where rows represent tuples and columns represent attributes or domains; connections between different data tables are established through shared attributes. Data access no longer relies on fixed paths but is achieved through logical relationships for joining and querying. This is the origin of the fundamental concepts of SQL (Structured Query Language).

The significance of relational databases lies not only in improving the efficiency of data management but also in providing a research framework centered on relationships. Within this framework, research subjects are no longer viewed as isolated entities but are understood within a broader network of connections. This approach aligns closely with the evolving trends in Digital Humanities and Quantitative Humanities Research. As Moretti (2013) points out, the focus of humanities research is gradually shifting from the examination of specific archives and individual objects to the analysis of patterns, structures, and relationships. Therefore, relational databases are not merely technical tools; they also provide an important methodological foundation for researchers to explore complex social relationships and historical connections.

Building on this foundation, network analysis can be viewed as a further extension of the logic of relational databases. In recent years, graph databases and network visualization have gradually emerged as essential tools for handling complex relational data. However, relevant research indicates that these two fields have long followed relatively independent development paths: graph database research primarily focuses on graph query languages, scalability, query optimization, and data management; while network visualization research places greater emphasis on graph layout, interaction design, visual encoding, and human comprehension. Despite these differing focuses, the two fields are in fact highly complementary. Database queries can rapidly generate a large number of paths and subgraphs, but presenting results solely in text form often leads to information overload. Therefore, researchers have proposed integrating schema discovery, interactive query exploration, query recommendation, and network visualization into a unified workflow, thereby achieving a deep integration of data management and network exploration (Klein et al., 2022).

This approach has since been implemented in several practices. For example, Neo4j (Neo4j inc., 2026) Bloom combines graph queries with interactive visualization within a single environment, allowing users to explore the database visually; GraphXR (Kineviz, 2025) emphasizes real-time interaction between queries and network exploration; Linkurious Enterprise (Linkurious, 2025) integrates search, relationship analysis, and network visualization into a unified workflow; and Graphistry (Graphistry Inc., 2025) utilizes GPU acceleration to enable real-time exploration of large-scale graph data. Together, these initiatives reflect a trend: network visualization is evolving from a tool for presenting query results into a vital component of the data exploration process.

However, most of the aforementioned practices are built on specialized graph database environments or existing databases. In contrast, data in digital humanities research is more commonly stored in SQL databases, spreadsheets, or CSV files. Easy Network is developed against this backdrop. Unlike existing graph database platforms, it does not rely on specialized graph database architectures but instead uses structured data and SQL queries created by users to directly generate network structures, providing researchers with a lightweight network exploration tool. By converting relationships in the database into visual networks, researchers can quickly switch between querying, exploring, and interpreting, thus more effectively identifying associative patterns and latent structures within the data.

I consider network visualization as an extension of the data management process rather than an analytical phase separate from the database. Relationships in the database are extracted, transformed, and reconstructed into networks through queries, and these networks further serve as important analytical tools for understanding historical relationships, social structures, and cultural connections. In this sense, Easy Network does not fill a gap in graph database technology itself, but rather bridges the gap between SQL queries and network exploration in a relational database environment, allowing network analysis to be more naturally integrated into the data management workflow of digital humanities research.

3. Easy Network

3.1. Overview of System Architecture

As the digitization of archaeological research advances, an increasing number of site surveys and excavation projects are generating large-scale artifact datasets. However, these data are typically stored in disparate locations, such as spreadsheets, databases, or different software platforms, resulting in the disconnect between data management, comparative analysis, and the presentation of results. Researchers often need to switch frequently between multiple tools, for example, using a database software to manage artifact information, statistical software to calculate similarity, and a network analysis platform for visualization. This workflow not only increases the cost of data processing but also reduces the reproducibility of research and the efficiency of analysis.

This study develops Easy Network, a platform aims to establish a comprehensive analytical environment that integrates data management, similarity analysis, and network visualization. Through a flexible attribute weighting mechanism, multi-level similarity calculations, and interactive network visualization features, Easy Network enables researchers to explore relationships between data at both the artifact and site levels, providing more efficient and intuitive technical support for comparative studies and network analysis in archaeology. This tool is divided into three modules based on functionality, as listed below and shown in the Figure 1.:

1) Inventory, which is used to import, store, and manage data of artifacts.

2) Similarity Analysis, based on data entries and specific algorithms;

3) Network Visualization that is traceable, adjustable, and flattens multidimensional data.

The fields in these three sections are linked via SQL to enable data retrieval, querying, and projection across different pages, as well as synchronous updates. Each module is explained in detail. For easier distinction, in the screenshots of the Easy Network interface below, the dark mode represents the artifact level, while the light mode represents the site level.

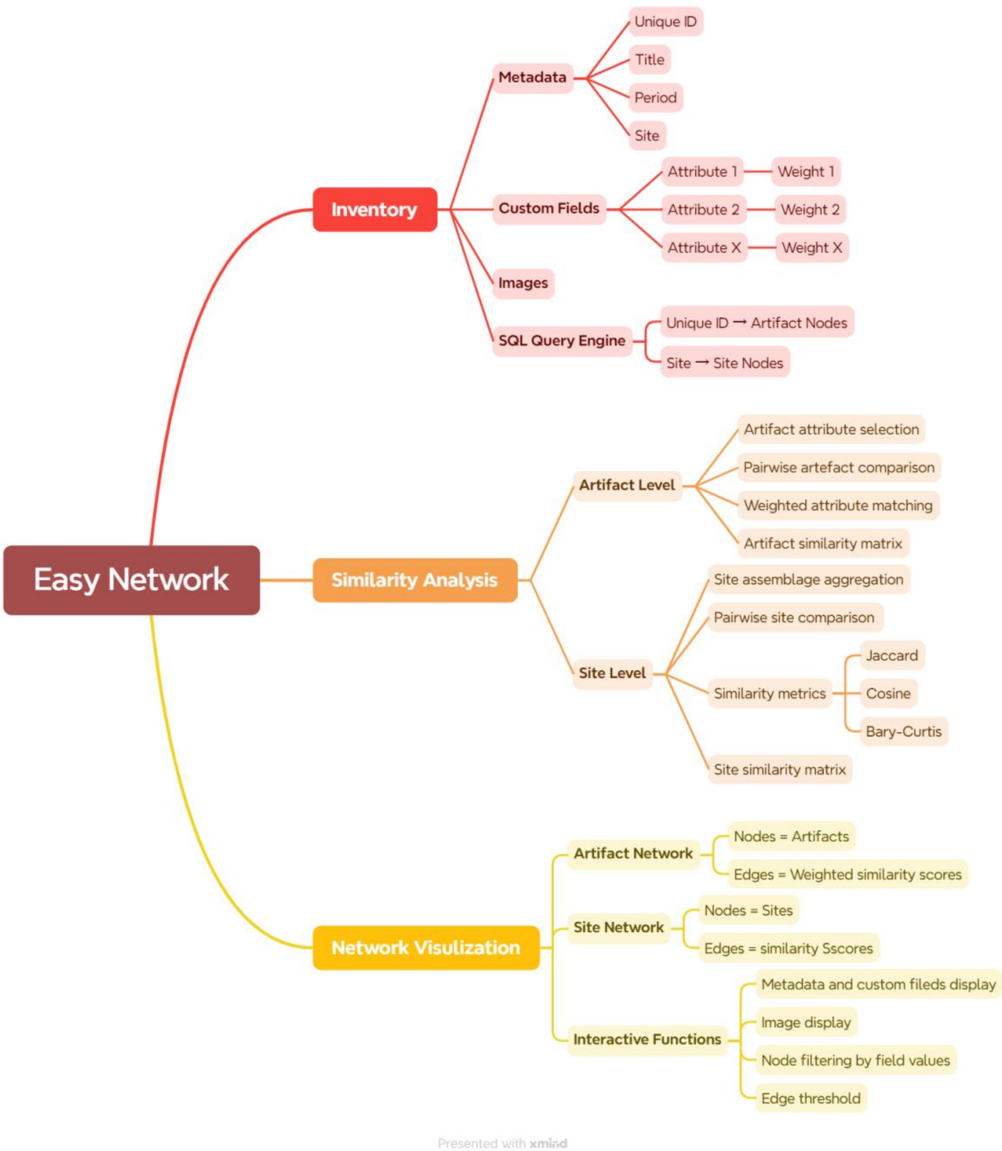

Figure 1. The demonstration of system architecture of Easy Network. The figure is created with Xmind (2025)

3.2. Inventory

Archaeological comparison depends heavily on how information is recorded. Differences in recording conventions, field names, and classificatory schemes often make it difficult to compare datasets produced by different projects. The Inventory module of Easy Network is therefore designed not simply as a storage system, but as a structured recording environment that promotes consistency, transparency, and interoperability across datasets. Similar to Tropy (2025), it supports the management of records and digital catalogues, while additionally providing bulk CSV import, tabular data management, and flexible field configuration that enables SQL-based integration with the Similarity Analysis and Network Visualization modules.

The Inventory page is divided into three functional areas. The left panel is the Global Field Control Panel, which allows users to batch import, generate, manage fields and use filters to select certain entries. This panel also allows users to merge two or more existing fields to create a new field; the resulting value is each value of the merged fields, linked by the sign "_". The middle area is the Workplace, the main space where users can import, organize, and view datasets, create subfolders for managing artifacts of different types, and switch between grid and list view modes. In list view, entries can be sorted in ascending or descending order based on different fields. For example, when sorted in ascending order by ID/Code, the entries are further sorted in ascending order by Title/Name, helping users view them in groups, as shown in the figure below.The right-hand panel, also called the Artifact Focus Panel, is used to view images and detailed information for the currently selected item, and allows users to edit the values of each field. Both the left and right panels can be collapsed to provide more space in the workspace.

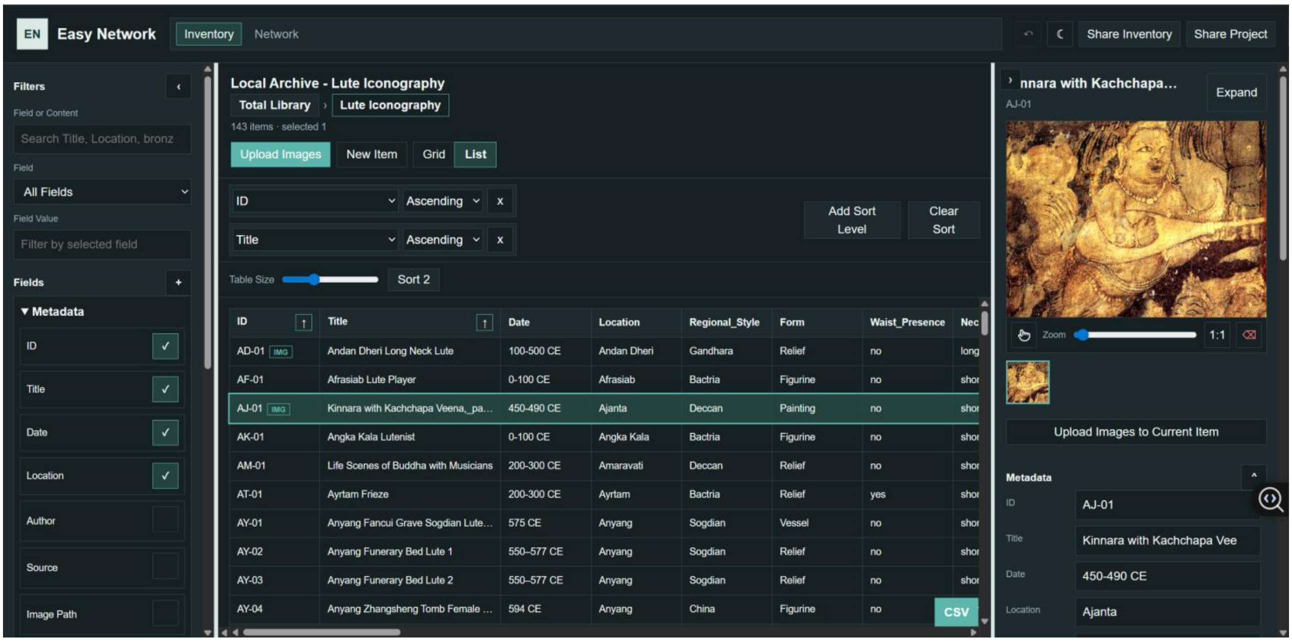

Figure 2. The interface of the Inventory module, list view

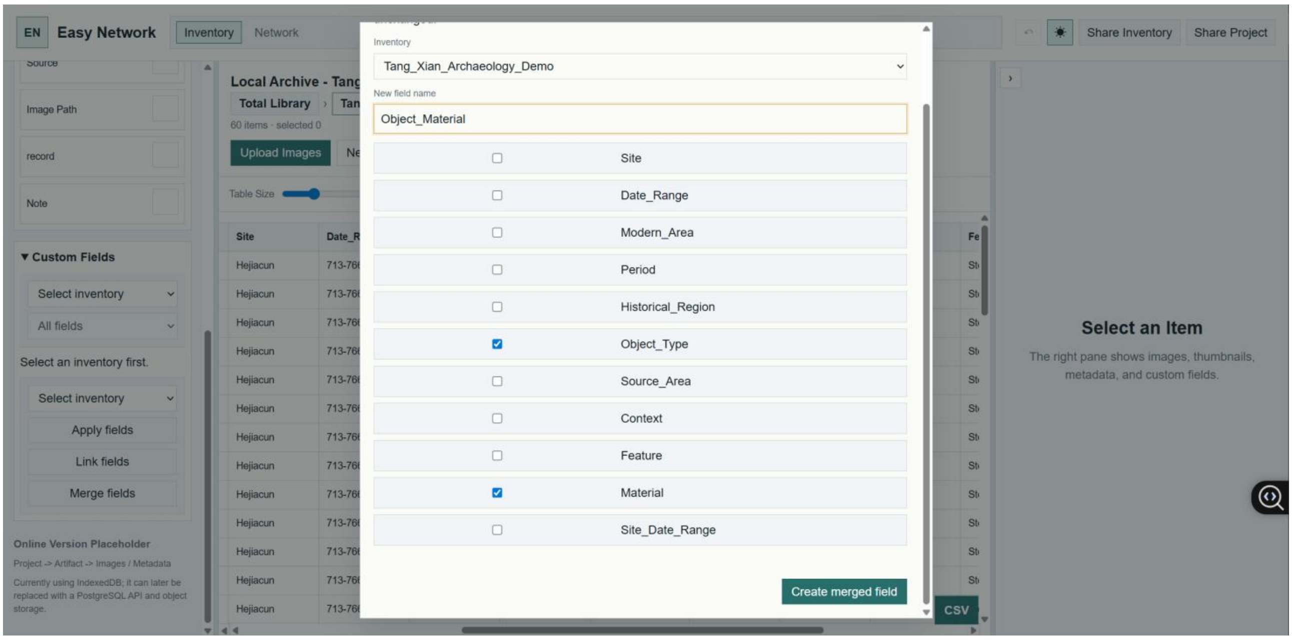

Figure 3. The interface of the Inventory module, the windows of merging fields.

The default fields in the “Metadata” include common artifact management fields such as the artifact’s unique ID, title (or name), date (or time), and location (or site). These can be used in the Similarity Analysis section to perform quantitative statistics across artifact types, enabling similarity analysis of excavated items based on shared sites and across different subfolders (sub-inventories) in the total inventory. For different sub-inventory, users can add, delete, or modify custom fields to serve specific research topics, and can enter values for each custom field in every record via the Artifact Focus panel on the right.

The Inventory feature primarily addresses two types of working scenarios. The first is for users who already have a dataset and simply need to import an existing CSV file. When importing, the system will ask whether to match fields with the Metadata fields already in the system, which sometimes have different names but the same semantic meaning, accommodating the conventions of different institutions and researchers and facilitates data integration. The second scenario involves users who only have raw data and need to create a dataset or digital inventory. This scenario better highlights the tool’s strengths: users can import records one by one, edit each field in detail, enter values, and create image directories for each record. The right-hand panel displays images and supports basic image editing to facilitate more detailed observation, with the ability to save edits. When users view an image, it hovers or expands in the central workspace, allowing them to enter data while observing.

3.3. Similarity Analysis

The purpose of Similarity Analysis is not merely to calculate numerical resemblance, but to provide a systematic way of exploring potential relationships among artifacts and sites. Archaeologists frequently infer cultural interaction, technological transmission, shared practices, or regional connections through similarities in material culture. However, such comparisons are often conducted qualitatively or rely on fixed typological categories. By transforming observable attributes or assemblage aggregation into comparable variables, Easy Network allows these relationships to be explored more explicitly and transparently.

The Similarity Analysis module organizes data recorded in the Inventory module and generates similarity matrices that can be directly translated into network visualizations. By integrating data management, comparison, and visualization within a single workflow, the module reduces the need for repeated data conversion between different software environments and keeps analytical decisions closely connected to the underlying data.

Similarity Analysis can be conducted at two levels: artifact level and site level. At the artifact level, objects are described through a series of observable variables defined by researchers or existing typological systems. Similarity is calculated by comparing the values of these variables across artifacts pairwisely. This approach follows recent studies of similarity network analysis in archaeology, such as Martin (2020) and van Heusden (2025), which decompose artifacts into comparable attributes and evaluate their degree of resemblance.

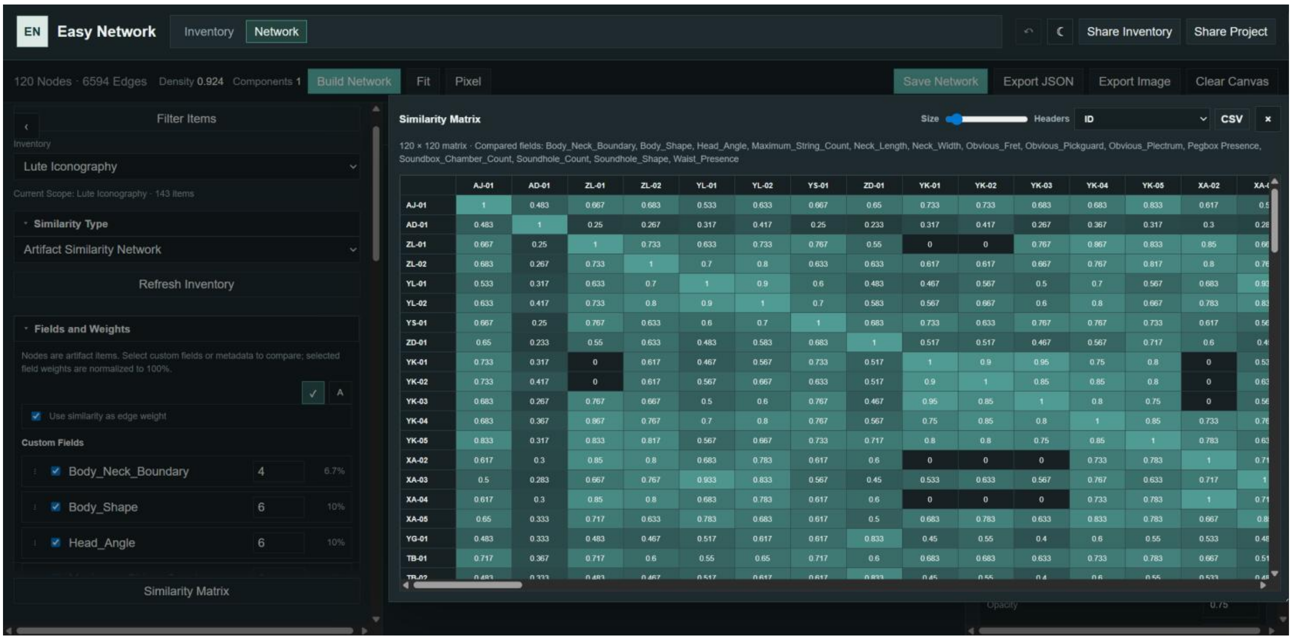

Figure 4. The interface of the Similarity Analysis module and matrix, artifact-level.

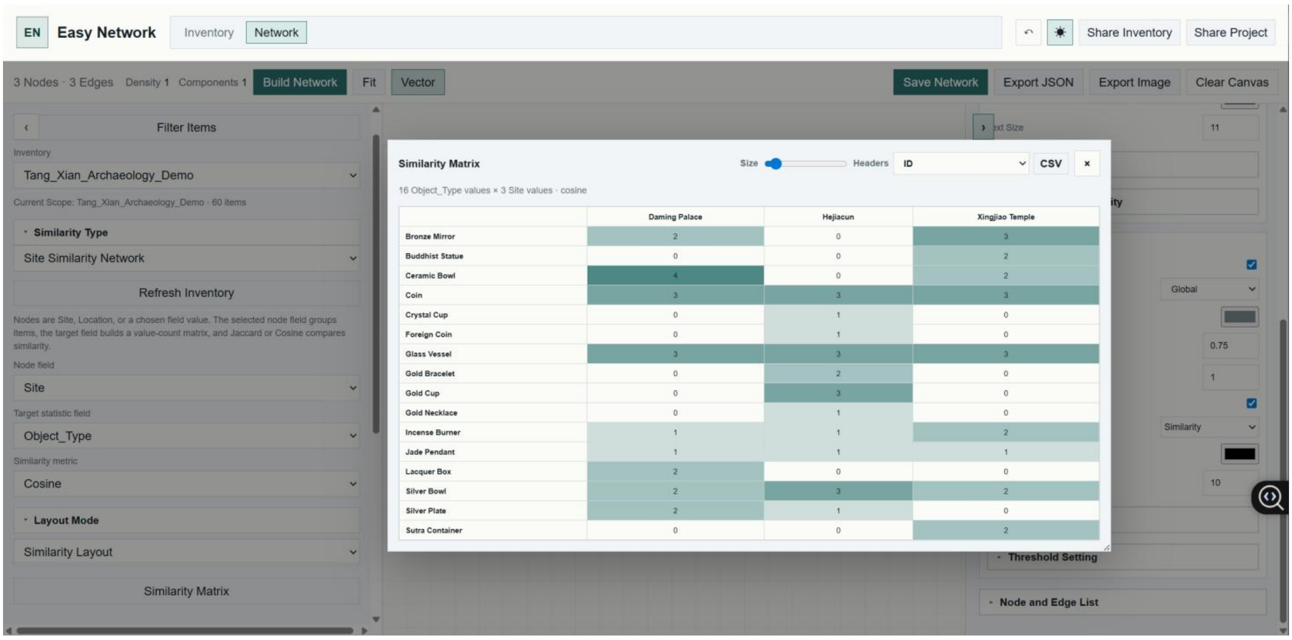

Figure 5. The interface of the Similarity Analysis module and matrix, site-level.

At the site level, each site is represented as an assemblage composed of different artifact categories and their abundances. Similarity is assessed by comparing the presence, frequency, or relative proportions of these categories across sites. The module supports several commonly used measures, including Jaccard Similarity (Jaccard, 1901), Bray-Curtis Dissimilarity (Bray & Curtis, 1957), and Cosine Similarity (Salton, 1975), allowing researchers to select methods appropriate to different datasets and research objectives.

The key feature of the module is the introduction of a flexible weighting system. Previous similarity analyses often assume that all variables contribute equally to similarity calculations. In archaeological research, however, different attributes may carry different interpretive significance. Functional attributes may be more relevant to technological practices, decorative attributes to stylistic traditions, and symbolic attributes to identity or ritual behaviour. Rather than treating similarity as a fixed property of archaeological data, Easy Network treats it as an analytical perspective. Different weighting schemes may produce different similarity networks, highlighting distinct dimensions of interaction, exchange, identity, or cultural affiliation. In this way, the module enables researchers not only to measure similarity, but also to explore how different interpretive assumptions shape the relationships revealed by the data.

3.4. Network Visualization

The Network Visualization module serves not merely as a means of presenting analytical results but as an exploratory tool for investigating complex archaeological relationships. Archaeological datasets often contain multidimensional information--including chronology, geography, typology, technology, and contextual associations--that can be stored and queried within databases but are difficult to comprehend simultaneously through tabular representations alone. By transforming similarity matrices into network graphs, researchers can more readily identify clusters, bridging nodes, spotting outliers, and comprehending broader patterns of association that may otherwise remain obscured within the original dataset.

To support transparent interpretation, each network node remains connected to its corresponding record in the Inventory module. When users hover over or select a node, the system displays associated images, metadata, and attribute values (custom fields) stored in the database. This enables researchers to move directly between network structures and the archaeological evidence from which they are derived, maintaining traceability between analytical results and source data. By integrating visualization with database exploration, the module reduces the risk of treating network patterns as detached abstractions and supports a more evidence-based interpretation of archaeological relationships.

The module provides adjustable edge filtering and graduated edge visualization. Large archaeological similarity networks often produce dense "hairball" graphs in which excessive numbers of weak connections obscure meaningful patterns. To address this issue, users can define threshold values and visualize gradients based on similarity scores, controlling which connections are displayed and how strongly they are emphasized. By progressively filtering weaker relationships and highlighting stronger similarities, researchers can focus on the most significant clusters and interaction patterns while still retaining the ability to explore the broader relational structure when necessary.

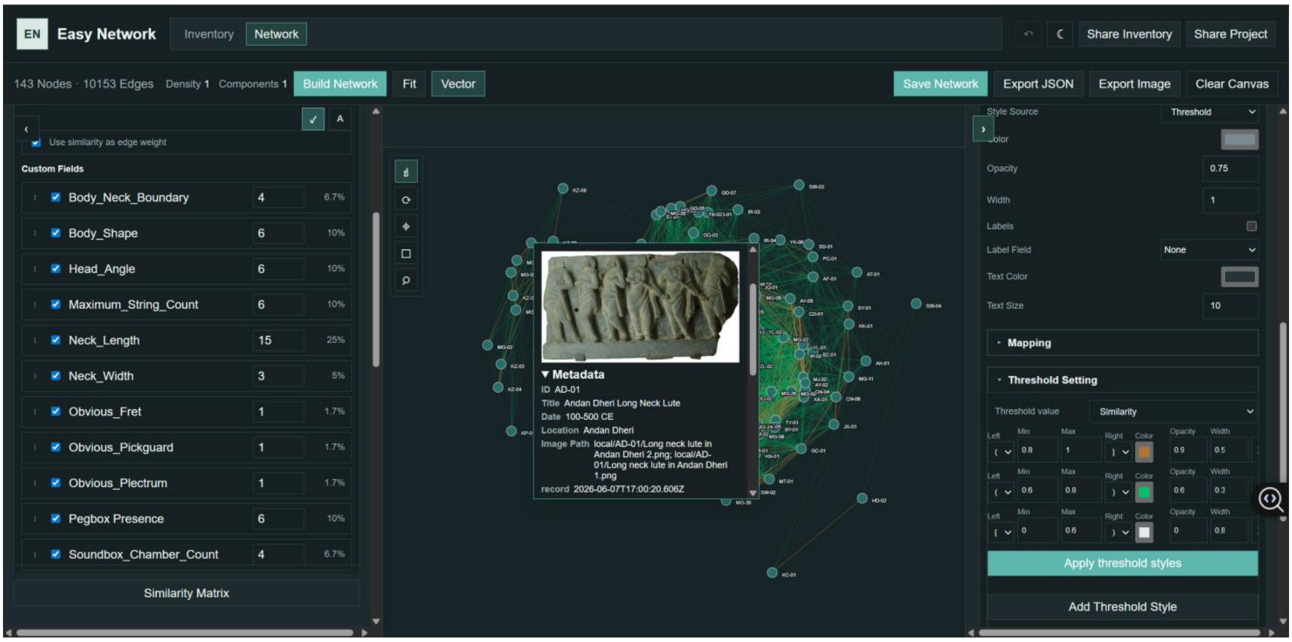

Figure 6. The interface of the Network Visualization module and network graph, artifact-level. Edges are graduated by three thresholds. Nodes could present the original data records.

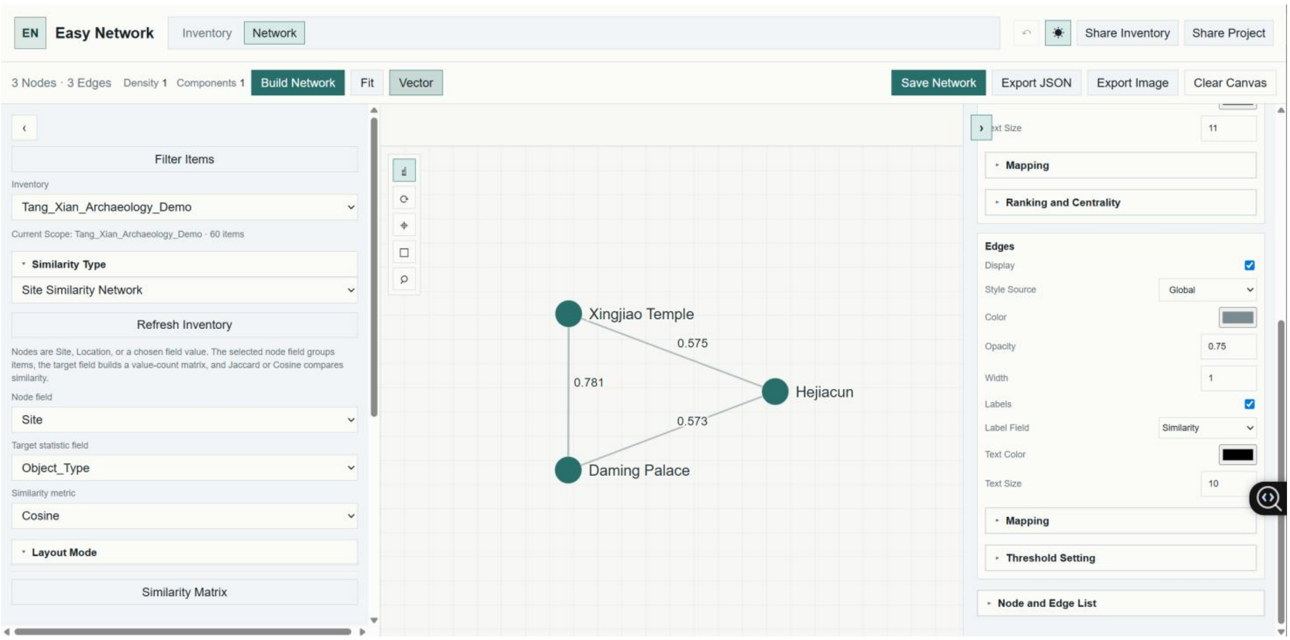

Figure 7. The interface of the Network Visualization module and network graph, site-level. Calculated by cosine similarity.

To facilitate the exploration of multidimensional datasets, the module also supports field-based visual encoding. Similar to the layer styling functions available in GIS platforms such as QGIS (QGIS Development Team, 2025), users can map selected database fields onto visual properties including node colour, shape, size, and labels etc. Different categories, periods, sites, or artifact attributes can be displayed simultaneously within the same network, enabling researchers to investigate how different variables intersect and structure observed patterns of similarity.

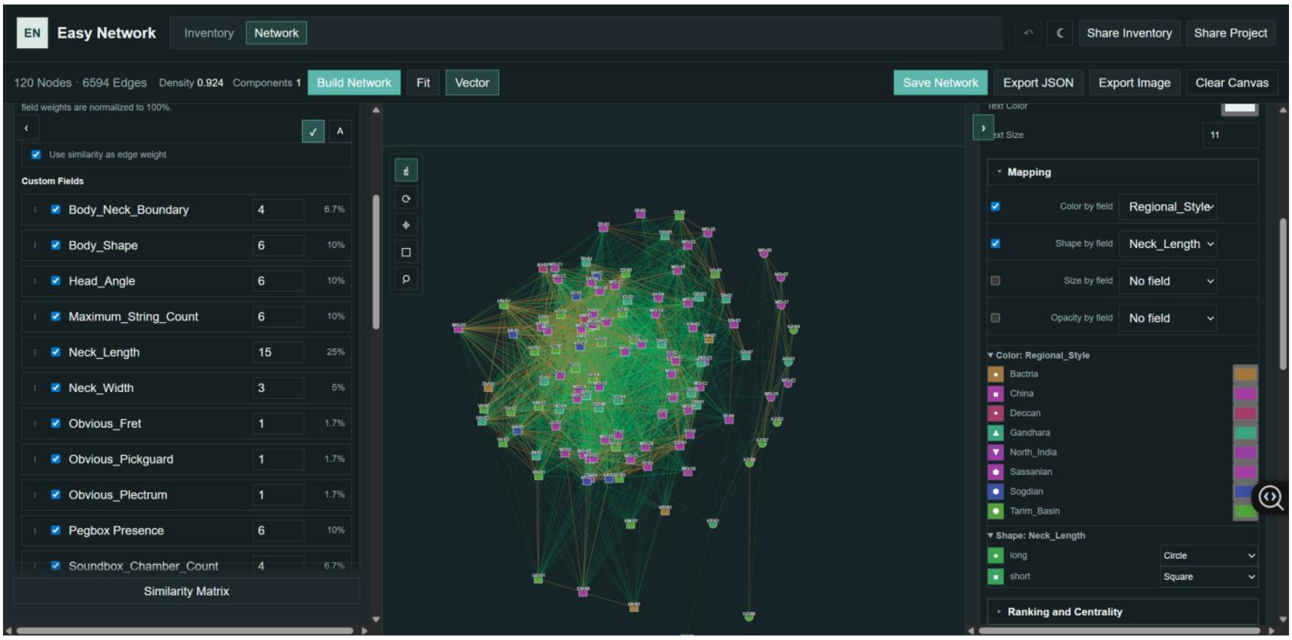

Figure 8. The demonstration of the Network Visualization module and network graph, mapping nodes with colors according to the field Regional_Style and shapes according to the field Neck_Length.

Because archaeological data are inherently temporal, the module further incorporates a chronological filtering system based on an interactive time slider. Users can dynamically display, remove, or compare nodes belonging to different chronological phases and observe how clusters, connections, and network structures change through time. This functionality allows the network to be explored not only as a static representation of relationships but also as a dynamic model through which long-term processes of cultural interaction, technological transmission, and regional development can be examined.

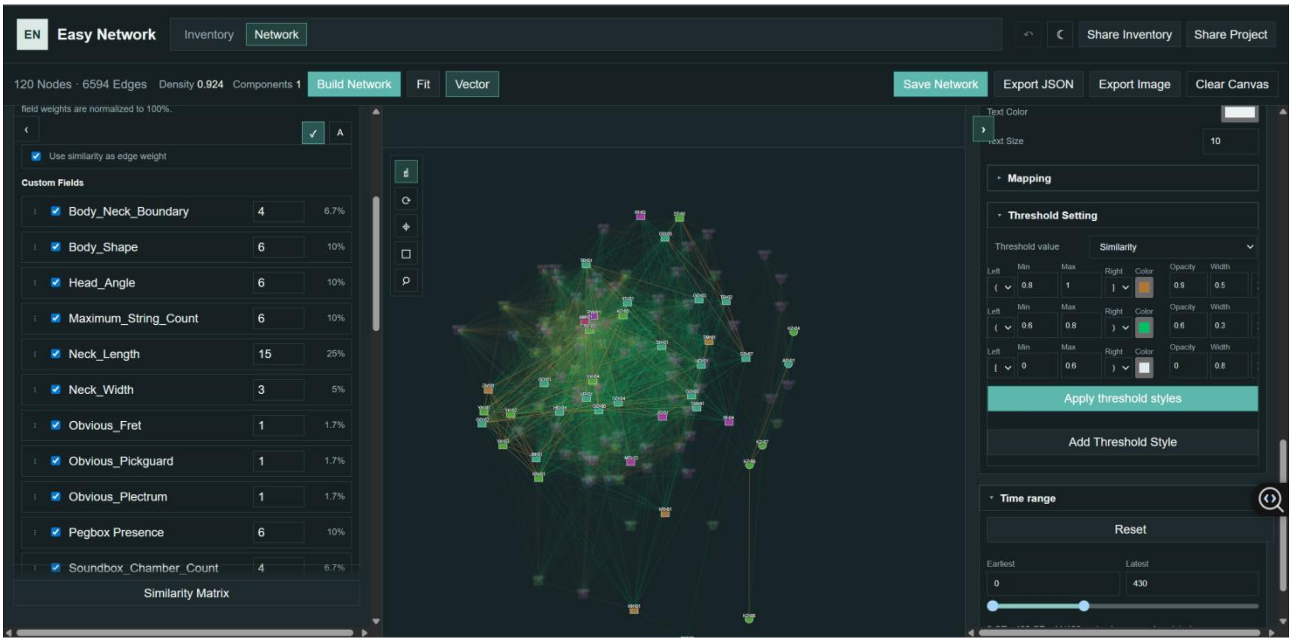

Figure 9. The demonstration of the Network Visualization module and network graph, filtered with the time range 0-430 CE. The nodes within the time range are focused.

In this sense, network visualization does not simply display analytical results. Rather, it functions as an exploratory interface between structured archaeological data and archaeological interpretation, enabling researchers to move continuously between database records, similarity calculations, and relational patterns within a single analytical environment.

4. Discussion

4.1. Network Analysis: Transforming Qualitative to Quantitative Data

In the humanities, data and visualization are often portrayed asc objective, neutral, and scientific. However, digital humanities scholar Johanna Drucker points out that quantification does not equal objectivity, because data in the humanities does not exist naturally but is selected, categorized, and organized during the research process (Drucker, 2011).

From this perspective, every field, category, and relationship in a database embodies the researcher's judgment. For example, which individuals are included in the scope of the study, which interactions are recorded as "relationships," and how social identities or functional categories are classified--these are not purely technical decisions, but interpretive ones. Therefore, when this data is further transformed into a network visualization, the graph does not present a completely objective reality, but rather a relational model constructed by the researcher based on specific research questions and classificatory logic.

Drucker (2011) argues that data visualization in the humanities should not be viewed as a transparent presentation of established facts, but rather understood as a process of interpretation. Unlike charts in the natural sciences that pursue precise measurement and certainty, the visualization of humanities data requires a greater emphasis on ambiguity, uncertainty, subjectivity, and relationality. In other words, the value of visualization lies not in proving a definitive conclusion, but in helping researchers discover patterns, pose questions, and make the interpretive process visible.

Within this framework, network analysis is not a technology for automatically generating objective knowledge, but rather a research method for exploring and organizing relationships. The nodes and edges displayed in a network graph are not direct projections of real-world relationships, but rather the result of selection and structuring within a database.

4.2. Defining Similarities: Paradox of Deconstruction and Reconstruction

Whether defining similarity at the artifact level based on matching values in certain attribute variables, or through statistical analysis of the absolute quantity or relative abundance of materials, artifacts, and other items at a site, both approaches rely on artificial classifications. These classifications constitute a categorization process; at the artifact level, this involves deconstruction, the definition of variables, and the selection of their values; at the site level, it involves the classification and definition of excavated artifacts and materials.

Adams & Adams (1991, pp. 31, 39-49, 91-95) argue that archaeology cannot avoid typology, yet is inevitably constrained by it. Types are not objective entities but analytical constructs composed of material, mental, and representational dimensions. All observation passes through human cognition, so artificial categorization is unavoidable; the key question is how categorization is conducted. Traditional typologies often impose fixed categories on material culture, despite the fact that artifacts frequently exhibit continuous variation, overlap, and hybridity. The approach of Easy Network treats artifacts as combinations of observable attributes rather than predefined types. Observable attributes are recorded first, and relationships among them are explored through comparison, statistical methods, and network analysis. Types are regarded as potential outcomes of analysis rather than its starting point. Accordingly, Easy Network serves not to reveal objective truth, but to provide a transparent and testable framework for visualizing and evaluating potential relationships.

4.3. Future Work

The innovation of Easy Network lies in its ability to reorganize data structures, filter and project existent information, and flatten multidimensional data. It is merely a research tool and does not generate new data, nor does it reveal any truths in an archaeological sense.

In future work, Easy Network will be refined to incorporate additional similarity calculation methods, supporting more sophisticated network analysis operations; I will explore three-dimensional projections by introducing time as a third axis to better observe changes in archaeological entities over time; and I will add a new functional panel where users can paste multiple observed nodes, facilitating comparative analysis by linking back to the original node information within the network graph. If this tool is published, I intend to make it accessible and shareable so that inventories, similarity matrices and network graphs, or entire projects can be shared and co-edited simultaneouslt by multiple users, thus fostering a new model of division of labor in archaeological research: whenever a researcher contributes an entry, the network analysis will automatically update and add the corresponding nodes and edges. I hope this will significantly improve work efficiency.

I have prepared demo datasets for readers. Hope you enjoy exploring this playground. Easy Network can be accessed by https://wenronghan.github.io/Easy-Network/. All the development files can be accessed by https://github.com/wenronghan/Easy-Network.

5. Technical Support

The website-based application developed for this study was implemented using HTML, CSS, and JavaScript. The front-end interface, artifact inventory management, CSV import functions, field editing tools, similarity calculations, and network visualisation components were developed through custom JavaScript modules. The application was deployed as a browser-based system and operated locally through a Node.js server for static file serving and local access management.

The development process was assisted by OpenAI Codex as an AI-supported coding environment. Network analysis and visualization functions were conceptually informed by established network-analysis frameworks, including NetworkX (Hagberg et al., 2008) and Gephi (Bastian et al., 2009), although these software packages were not directly integrated into the final application.

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